A Four-Stage Quantitative Research Pipeline: Signal Discovery, Strategy Construction, Portfolio Construction, and Risk Analysis

Adekanmi Daniel Omole

Email: adekanmiomole@gmail.com

GitHub: github.com/Ade-omole/Quant-research-pipeline

LinkedIn: linkedin.com/in/adekanmi-daniel-omole

Edited: October 28, 2026

Project Overview

Many quantitative projects stop once a model looks good on historical data. This project asks a more practical question: **which signals still matter after they are turned into trades, combined into a portfolio, and charged realistic costs?**

It uses daily prices of eight ETFs (SPY, TLT, GLD, USO, UUP, EFA, EEM, FXE) from 2004 to August 2026, a period that includes the 2008 crisis, the 2020 COVID crash and the 2022 inflation shock. Three strategies are built (volatility timing, regime switching and adaptive asset allocation, AAA), combined into portfolios, and then tested with sensitivity checks, historical stress episodes and a Monte Carlo simulation. **All strategy and portfolio results are walk-forward out of sample** (July 2006 to August 2026) and net of costs.

The main result is that adding AAA, a cross-asset momentum strategy, to a portfolio of two SPY-based strategies raised the portfolio Sharpe ratio from 0.77 to 0.88 and reduced its maximum drawdown from -23.5% to -16.8%, mainly because AAA is much less correlated with the other two. Simple weighting schemes did at least as well as mean–variance optimization. The full report contains the methods, appendices and supporting tables.

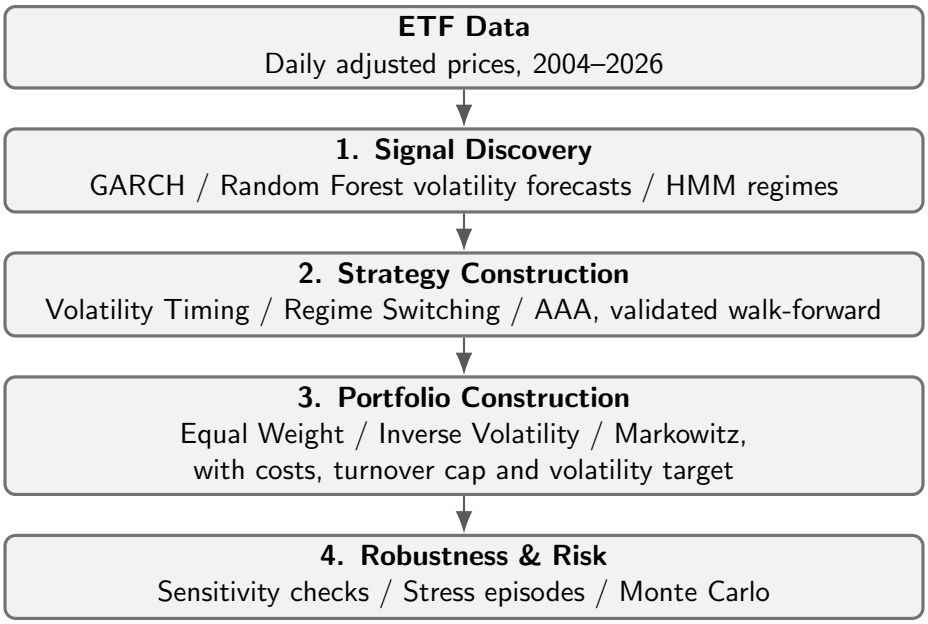


Figure 1: Project workflow.

1 Research Design

Each stage answers one question:

1. **Is there a signal?** Can SPY volatility be forecast, and can market regimes be identified?
2. **Can it be traded?** Do rules built on these signals hold up out of sample after costs?
3. **Does it improve a portfolio?** Does combining strategies help after turnover limits, costs and a volatility target?
4. **How sensitive is the result?** To sub-periods, cost levels, parameters, crises and resampled histories.

Research block	Question tested	Output used later
Volatility forecasting	Can next-period SPY risk be forecast from past volatility and returns?	Volatility timing: scale SPY exposure by forecast risk.
HMM regimes	Can SPY's history be split into calm, intermediate and stress regimes?	Regime switching: reduce SPY exposure in weaker regimes.
Cross-asset momentum	Can the rest of the ETF universe be used, not only SPY?	AAA: hold the strongest ETFs, weighted by inverse volatility.
Portfolio allocation	Does combining the strategies help after costs and constraints?	Equal, inverse-volatility and Markowitz portfolios.
Robustness and risk	Does the result depend on one period, cost level or parameter?	Sensitivity tables, stress episodes, Monte Carlo.

Table 1: Research design.

Positions are lagged one day, every unit of turnover costs 7 basis points (5 transaction cost plus 2 slippage), and portfolios use a 0.35 turnover cap and a 10% volatility target. Models are refit on expanding training windows (241 monthly folds), and AAA's two parameters are chosen on training data only. SPY buy-and-hold and a 60/40 SPY/TLT portfolio are the benchmarks.

2 Findings

Volatility forecasts were useful for risk management

Both GARCH and a random forest tracked realized volatility on the test period, with smaller errors for the random forest. This is a risk forecast, not a forecast of market direction, and the high correlation partly reflects that volatility is persistent: tomorrow's 21-day realized volatility shares 20 of its 21 days with today's.

Model	MSE	MAE	Correlation
Random Forest	0.000255	0.008369	0.990378
GARCH$(1,1)$	0.001395	0.022807	0.945221

Table 2: Volatility forecasts on the test period, in annualized volatility units.

The three strategies

Volatility timing holds more SPY when forecast risk is low and less when it is high. **Regime switching** holds 100%, 35% or 0% SPY depending on the HMM regime. **AAA** ranks the eight

ETFs each month by past return, keeps the strongest, weights them by inverse volatility, and moves any holding with negative momentum into Treasuries (TLT).

Metric	VolTiming	RegimeHMM	AAA	SPY	60/40
Sharpe ratio	0.81	0.75	0.83	0.65	0.82
Sortino ratio	1.05	0.83	1.10	0.80	1.06
CAGR (%)	10.68	6.18	8.59	11.35	9.00
Max drawdown (%)	-30.18	-14.95	-17.32	-55.19	-29.92
Calmar ratio	0.35	0.41	0.50	0.21	0.30
Volatility (%)	13.69	8.52	10.61	19.38	11.33
VaR_{95} (%)	1.45	0.83	1.06	1.79	1.06
ES_{95} (%)	2.12	1.40	1.61	2.98	1.68
Turnover (x/yr)	6.81	11.42	6.78	–	–

Table 3: Strategy results, walk-forward out of sample, after costs, July 2006 to August 2026.

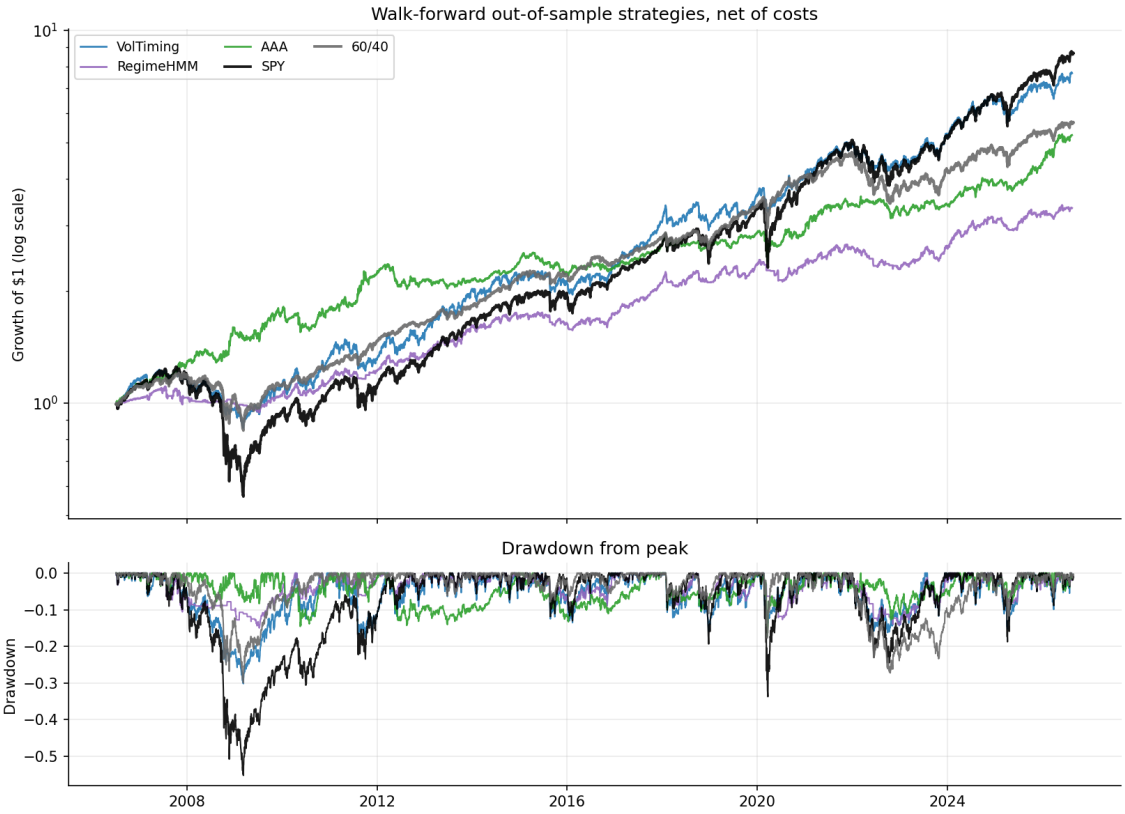


Figure 2: Out-of-sample equity and drawdown curves, net of costs.

All three strategies had a higher Sharpe ratio than SPY (0.75 to 0.83 against 0.65) and much smaller drawdowns, at the cost of lower growth. Their Sharpe ratios are close to each other and

to 60/40, so the main differences are in drawdown: -30% for volatility timing, -15% for regime switching and -17% for AAA, against -55% for SPY.

3 Portfolio Construction

Two menus are compared: **Orig2** (volatility timing and regime switching) and **Core3** (Orig2 plus AAA). The two SPY-based strategies correlate at 0.84, while AAA correlates about 0.4 with each, so AAA is the only one that adds real diversification.

Menu	Equal weight	Inverse vol	Markowitz	Markowitz regularized
Orig2	0.79	0.77	0.72	0.72
Core3	0.96	0.88	0.82	0.82

Table 4: Net Sharpe ratio by menu and weighting method, walk-forward and after all costs.

Metric	Orig2 inv-vol	Orig2 equal	Core3 inv-vol	Core3 equal	Core3 Mkw. reg.	SPY	60/40
Sharpe ratio	0.77	0.79	0.88	0.96	0.82	0.64	0.80
Sortino ratio	0.94	0.98	1.10	1.22	1.05	0.79	1.04
CAGR (%)	7.79	8.00	9.09	10.03	8.40	11.22	8.80
Max drawdown (%)	-23.52	-23.76	-16.79	-15.97	-21.19	-55.19	-29.92
Calmar ratio	0.33	0.34	0.54	0.63	0.40	0.20	0.29
Volatility (%)	10.46	10.47	10.50	10.59	10.58	19.47	11.38
VaR_{95} (%)	1.03	1.05	1.05	1.07	1.09	1.80	1.07
ES_{95} (%)	1.68	1.65	1.66	1.63	1.62	2.99	1.69

Table 5: Key portfolios in detail, with SPY and 60/40 over the same window.

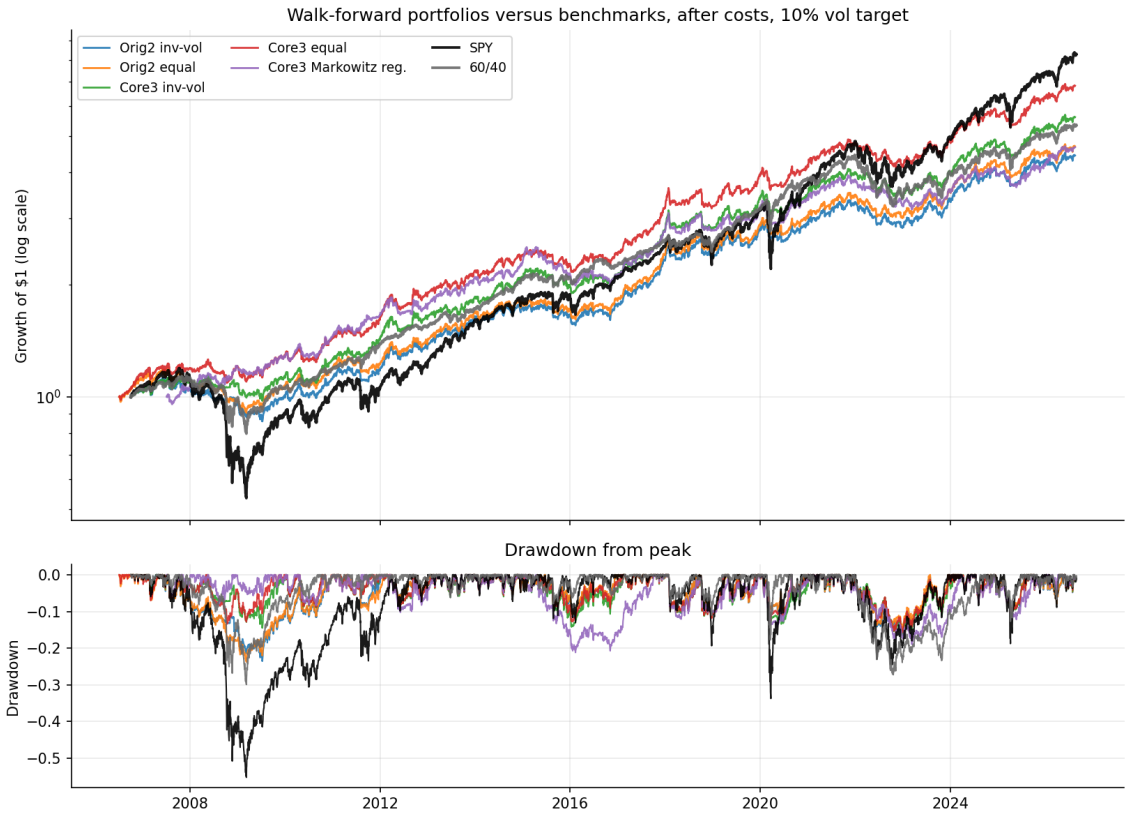


Figure 3: Portfolio equity and drawdown curves after turnover cap, costs and volatility targeting.

Core3 had a higher Sharpe ratio than Orig2 under every weighting method, and a clearly smaller drawdown under equal and inverse-volatility weighting. Equal weighting (0.96) and inverse volatility (0.88) did better than Markowitz (0.82), likely because expected returns estimated from a year of data are noisy. Compared with SPY, Core3 gave up about two percentage points of annual growth for a much smaller maximum drawdown (-16.8% against -55.2%).

4 Robustness and Risk

- **Sub-periods:** AAA and Core3 had positive Sharpe ratios in all five sub-periods; Core3 was at or above SPY in each, but below 60/40 in 2010–21.
- **Costs:** at 30 bps per unit of turnover, AAA still earned 0.68, slightly above SPY; Core3, which pays costs twice, fell to 0.57, below SPY.
- **Parameters:** 15 of 20 fixed AAA settings beat SPY, and replacing TLT with GLD as the defensive asset changed the Sharpe ratio by less than 0.01.

Stress episodes. Each strategy's out-of-sample returns are measured over nine historical crisis windows. In 2008 SPY lost 54% and 60/40 lost 29%, while Core3 lost 10% and AAA gained

24%. The weak spots were short, sudden shocks (February 2018, the 2025 tariff episode) and 2022, when stocks and bonds fell together.

Episode	VolTiming	RegimeHMM	AAA	Core3 inv-vol	SPY	60/40
2008 global financial crisis	-28.8	-8.8	24.2	-9.9	-54.0	-28.6
2010 flash crash / euro scare	-11.5	-6.6	-7.4	-7.9	-14.7	-3.4
2011 US downgrade / euro crisis	-13.2	-3.0	21.9	1.3	-16.3	1.8
2015 China deval / Aug flash crash	-9.2	-5.6	-0.7	-4.9	-8.5	-4.5
2018 Feb VIX spike	-6.7	-5.0	-5.3	-6.8	-4.1	-4.0
2018 Q4 rate-hike selloff	-12.5	-6.5	-1.3	-8.7	-13.5	-6.5
2020 COVID crash	-10.5	-5.3	-6.1	-8.8	-17.2	-3.1
2022 inflation / hiking cycle	-11.8	-10.7	-6.3	-12.3	-17.7	-24.0
2025 tariff shock	-4.0	-4.7	-3.3	-4.8	-0.5	-2.3

Table 6: Total return (%) through each stress episode, walk-forward and net of costs.

Monte Carlo. 5,000 alternative one-year paths are built by resampling blocks of each portfolio's own daily returns. Core3's median simulated year returned 9.1% against 12.9% for SPY, but its 5th-percentile year lost 7.3% against 17.7%: less upside and less downside.

Portfolio	Median 1y (%)	5th pct 1y (%)	95th pct 1y (%)	P(loss) (%)	MC ES_{95} (%)	Median max DD (%)
Orig2 inv-vol	7.73	-8.43	26.57	22.20	11.99	-7.94
Orig2 equal	8.12	-8.23	26.12	21.82	11.91	-8.07
Core3 inv-vol	9.07	-7.32	28.04	18.94	11.25	-7.80
Core3 equal	10.06	-7.06	28.59	17.02	10.72	-7.68
Core3 Markowitz reg.	8.66	-8.27	27.24	21.42	12.26	-7.84
AAA	8.45	-6.91	26.30	18.00	10.68	-7.26
SPY	12.89	-17.65	42.02	23.36	25.43	-13.25
60/40	9.62	-9.05	27.00	18.64	13.79	-8.06

Table 7: Distribution of one-year outcomes across 5,000 block-bootstrap paths.

5 Limitations

The results come from one historical sample of eight ETFs. Costs are a fixed assumption, without market impact or wider spreads in stressed markets. Parameters were chosen without hindsight, but the choice of strategy family was informed by the momentum literature. Treasuries are the only defensive asset, which is why 2022 was difficult. Several differences in the tables are small compared with the uncertainty of a Sharpe ratio estimated over 20 years.

6 Conclusion

The project takes ideas from signal research to a costed, walk-forward tested portfolio. Volatility and regime signals helped control risk, and adding a cross-asset momentum strategy improved the portfolio mainly through diversification. Simple weighting was at least as good as optimization. The stress tests and Monte Carlo point the same way: the portfolios gave up some return relative to SPY in exchange for considerably smaller losses in bad periods.